

Exp 1: Caesar Cipher

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[100], k, i;
```

```
    printf("Enter text: ");
```

```
    scanf("%s", str);
```

```
    printf("Enter key: ");
```

```
    scanf("%d", &k);
```

```
    for(i = 0; str[i] != '\0'; i++) {
```

```
        if(str[i] >= 'A' && str[i] <= 'Z')
```

```
            str[i] = (str[i] - 'A' + k) % 26 + 'A';
```

```
        else if(str[i] >= 'a' && str[i] <= 'z')
```

```
            str[i] = (str[i] - 'a' + k) % 26 + 'a';
```

```
    }
```

```
    printf("Encrypted text: %s\n", str);
```

```
    for(i = 0; str[i] != '\0'; i++) {
```

```
        if(str[i] >= 'A' && str[i] <= 'Z')
```

```
            str[i] = (str[i] - 'A' - k + 26) % 26 + 'A';
```

```
        else if(str[i] >= 'a' && str[i] <= 'z')
```

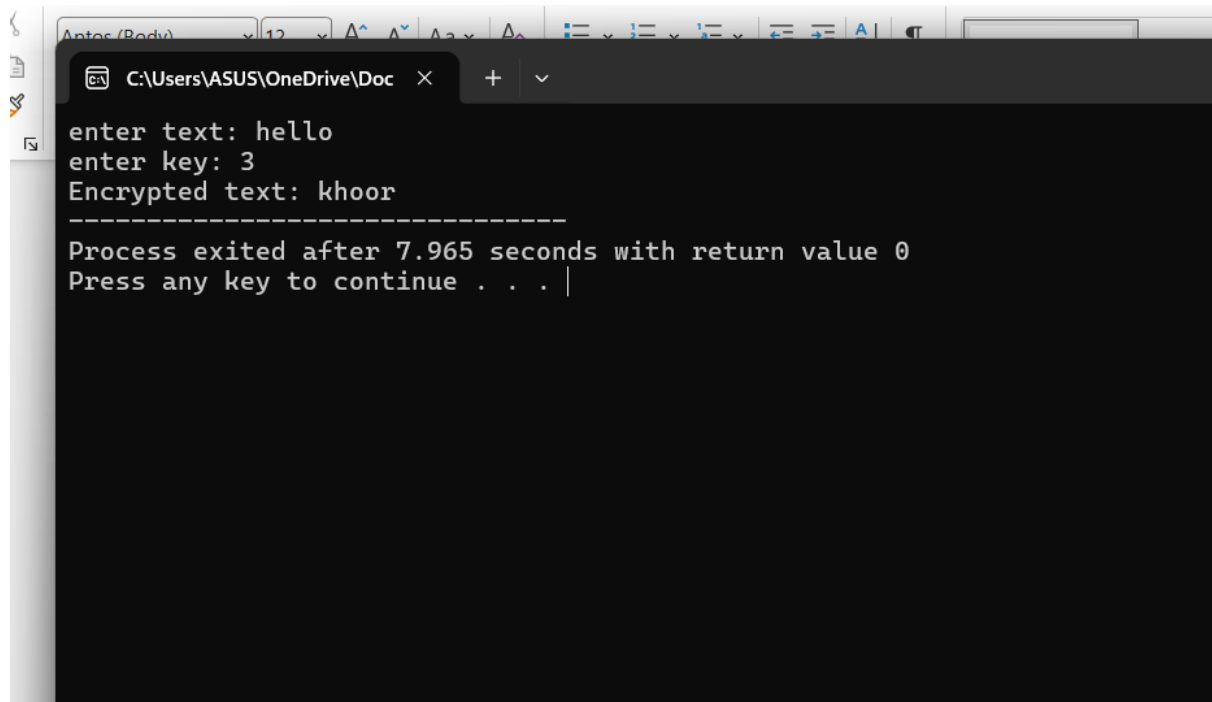
```
            str[i] = (str[i] - 'a' - k + 26) % 26 + 'a';
```

```
    }
```

```
printf("Decrypted text: %s\n", str);
```

```
return 0;
```

```
}
```



The screenshot shows a C++ IDE with a single tab titled "C:\Users\ASUS\OneDrive\Doc". The code editor contains the following text:

```
enter text: hello
enter key: 3
Encrypted text: khood
-----
Process exited after 7.965 seconds with return value 0
Press any key to continue . . . |
```

The output shows that the input text "hello" and key "3" were used to produce the encrypted text "khood". The program then displays a separator line and a message indicating it exited after 7.965 seconds with a return value of 0, followed by a prompt to press any key to continue.